

Hawkeye EQMS · Design Control (medical device) · Feature Guide

Pharma-EQMS comparison · click-by-click guide · personas + lifecycle walkthrough · AI map · test results.

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1. Module overview

Module	Design Control (medical device)
Pharma purpose	Track a medical-device design through 8 phases (Planning → Input → Output → Review → Verification → Validation → Transfer → Changes). Maintain the Design History File (DHF). Link to Risk, Change Control, CAPA.
Compliance basis	21 CFR 820.30 (Design Controls) · ISO 13485:2016 §7.3 · ISO 14971 (risk management) · MDR 2017/745
Hawkeye routes	/design-controls (frontend) · /api/design-controls (backend)
Hawkeye model file	backend/src/models/DesignControlModel.js
Tenant module flag	modules.DESIGN_CONTROL

Designs follow DRAFT → ACTIVE → DESIGN_FREEZE → TRANSFERRED → OBSOLETE. The 8 design phases (PLANNING → INPUT → OUTPUT → REVIEW → VERIFICATION → VALIDATION → TRANSFER → CHANGES) progress sequentially via /advance-phase. Each phase status: NOT_STARTED → IN_PROGRESS → COMPLETED → BLOCKED.

2. Standard pharma EQMS vs Hawkeye — gap analysis

Each row is a capability a pharma QA team would expect from an EQMS deviation module. Hawkeye column shows what's actually shipped. **MET** = parity, **PARTIAL** = ships but with caveats, **GAP** = not implemented yet.

Pharma expectation	Hawkeye implementation	Status	Note
8-phase design lifecycle per 21 CFR 820.30 21 CFR 820.30	phaseKey enum with 8 phases · sequential transition rule.	MET	
Device classification (Class I / II / III / IVD) 21 CFR 860	deviceClass enum · 4 values.	MET	
Regulatory pathway (510K / PMA / De Novo / CE Mark) 21 CFR 807, 814	regulatoryPathway enum · 5 values.	MET	
Design Inputs (Requirements) with traceability to outputs 21 CFR 820.30(c)	designInputs[] subdocument with requirementId + source + priority + verified + traceableToOutput.	MET	
Design Outputs (Specifications) traceable to inputs 21 CFR 820.30(d)	designOutputs[] with outputType + documentRef + tracedToInput.	MET	
Design Reviews with attendees + decision (PROCEED / REVISE / HOLD) 21 CFR 820.30(e)	designReviews[] subdocument.	MET	
Verification (V&V protocols + reports) 21 CFR 820.30(f)	verificationProtocolRef + verificationReportRef fields.	MET	
Validation (V&V protocols + reports) 21 CFR 820.30(g)	validationProtocolRef + validationReportRef fields.	MET	
Design Transfer with manufacturing site link 21 CFR 820.30(h)	POST /:id/transfer with manufacturingSiteId.	MET	
Design History File (DHF) 21 CFR 820.30(j)	dhfRef field linking to controlled-document repository.	MET	
Linked Risk / Change Control / CAPA ISO 14971	linkedRiskItems[] + linkedChangeControlIds[] + linkedCapalds[].	MET	
Auto-numbered (DC-YYYY-NNNN) GMP traceability	Pre-save generator.	MET	

3. Personas + role mapping

Login password for every persona: `EqmsDemo@2026` . Roles drive which API endpoints respond — see Hawkeye `roleMiddleware.permit()` .

Persona	Email	Responsibilities in this module	Lifecycle states they touch
Product Engineer (custom role) engineer	(engineer-side login)	Authors design, captures inputs + outputs, advances phases.	DRAFT ACTIVE
Head of QA James admin · reviewer	qa.head@novex-pharma.demo	Reviews phase artifacts, signs off design reviews, approves verification.	ACTIVE (REVIEW + VERIFICATION)
Marcus Brown Reg Affairs	regulatory@novex-pharma.demo	Verifies regulatory pathway, approves transfer.	ACTIVE (TRANSFER)
Manufacturing (Production Head Michael) production	production.head@novex-pharma.demo	Receives transferred design, accesses DHF.	TRANSFERRED

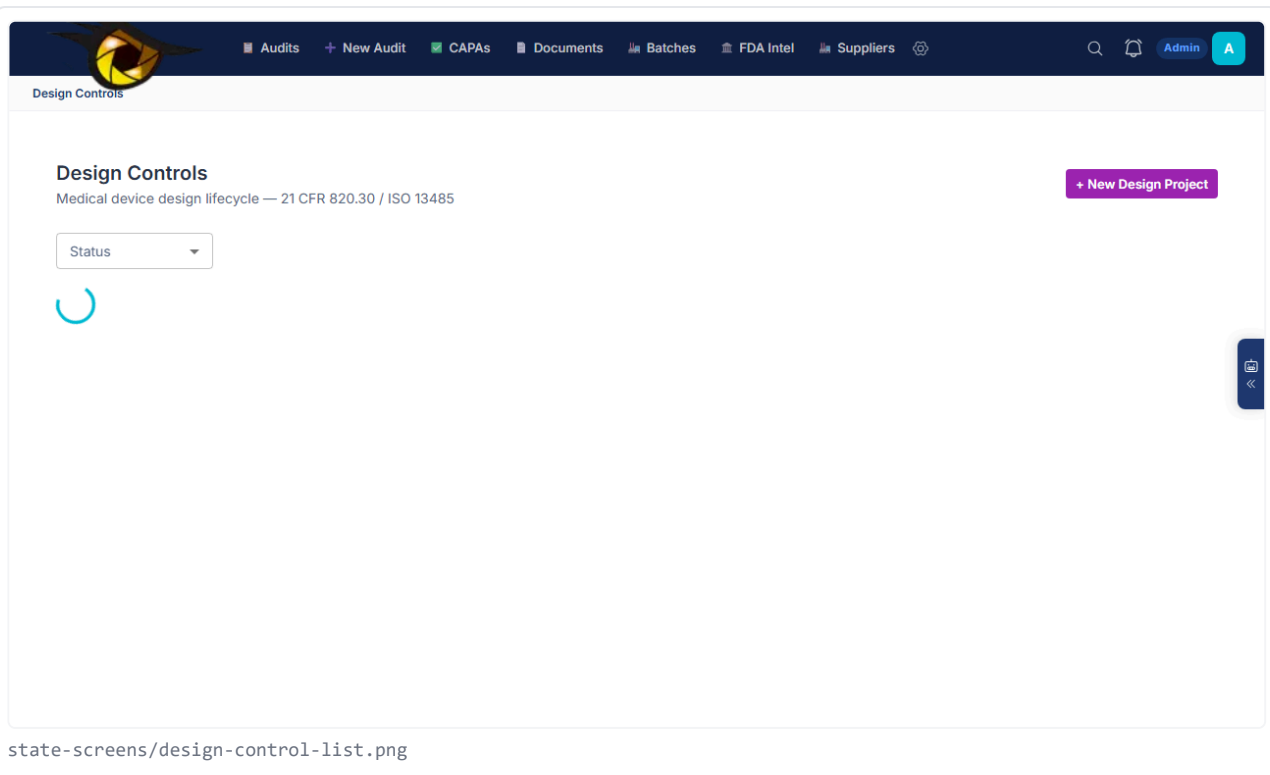
4. Feature catalogue (click-by-click)

4.1 Design register

What it does	Lists all designs with deviceClass + regulatoryPathway + currentPhase + status.
Menu / route	/design-controls
Available to roles	any tenant viewer
Backend API	GET /api/design-controls

CLICK-BY-CLICK

- NAVIGATE** Click 'Design Controls' (admin nav)
→ Page renders



4.2 + Start Design dialog

What it does	Author a new design.
Menu / route	/design-controls · top-right '+ Start Design'
Available to roles	engineer · admin
Backend API	POST /api/design-controls

CLICK-BY-CLICK

- CLICK** Click '+ Start Design'
→ Dialog opens
- TYPE** Fill title + productName
→ required
- CLICK** Pick deviceClass (CLASS_I / II / III / IVD)
→ required
- CLICK** Pick regulatoryPathway (510K / PMA / DE_NOVO / CE_MARK / OTHER)
→ required
- CLICK** Click 'Save'
→ Status=DRAFT · designNumber=DC-YYYY-NNNN · 8 phases initialized to NOT_STARTED

FIELDS CAPTURED

Field	Required	Validation / values	Notes
title	Yes	string	
productName	Yes	string	

Field	Required	Validation / values	Notes
deviceClass	Yes	CLASS_I CLASS_II CLASS_III IVD	
regulatoryPathway	Yes	510K PMA DE_NOVO CE_MARK OTHER	
riskLevel	No	LOW MEDIUM HIGH UNACCEPTABLE	

4.3 Advance phase

What it does	Marks current phase COMPLETED + next phase IN_PROGRESS. Auto-flips overall status when last phase done.
Menu / route	Design detail · 'Advance to next phase' button
Available to roles	engineer · QA
Backend API	POST /api/design-controls/:id/advance-phase

CLICK-BY-CLICK

- CLICK** Click 'Advance to next phase'
 - Phase N → COMPLETED · Phase N+1 → IN_PROGRESS · status auto-flips DRAFT → ACTIVE on first advance

4.4 Add Design Review

What it does	Capture a design review meeting with attendees + decision (PROCEED / REVISE / HOLD).
Menu / route	Design detail · 'Add Review' button (visible from REVIEW phase onwards)
Available to roles	QA · reviewer
Backend API	POST /api/design-controls/:id/reviews

CLICK-BY-CLICK

- CLICK** Click 'Add Review'
 - Drawer opens
- CLICK** Pick reviewDate + reviewPhase
 - required
- CLICK** Pick attendees[]
 - ObjectIds[]
- CLICK** Pick decision (PROCEED / REVISE / HOLD)
 - required
- TYPE** Fill actionItems[] + minutesRef
- CLICK** Click 'Save'
 - designReviews[] += entry

4.5 Add Design Input / Output

What it does	Capture requirements (inputs) + specifications (outputs) with traceability.
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Menu / route	Design detail · 'Inputs' / 'Outputs' tabs
Available to roles	engineer
Backend API	PUT /api/design-controls/:id (with new entries)

CLICK-BY-CLICK

- 1. **CLICK** Open Inputs tab · '+ Add Input'
→ Form opens
- 2. **TYPE** Fill requirementId + description
→ required
- 3. **CLICK** Pick source (USER_NEED / REGULATORY / STANDARD / RISK / BUSINESS)
- 4. **CLICK** Pick priority (CRITICAL / HIGH / MEDIUM / LOW)
- 5. **CLICK** Save
→ designInputs[] += entry

4.6 Design Transfer (TRANSFER phase)

What it does	Formally transfer design to manufacturing site.
Menu / route	Design detail · 'Transfer' button (visible in TRANSFER phase)
Available to roles	QA · RA · admin
Backend API	POST /api/design-controls/:id/transfer

CLICK-BY-CLICK

- 1. **CLICK** Click 'Transfer'
→ Drawer opens
- 2. **CLICK** Pick manufacturingSiteId
→ required
- 3. **CLICK** Click 'Confirm transfer'
→ Status flips DRAFT/ACTIVE → DESIGN_FREEZE → TRANSFERRED · currentPhase=CHANGES

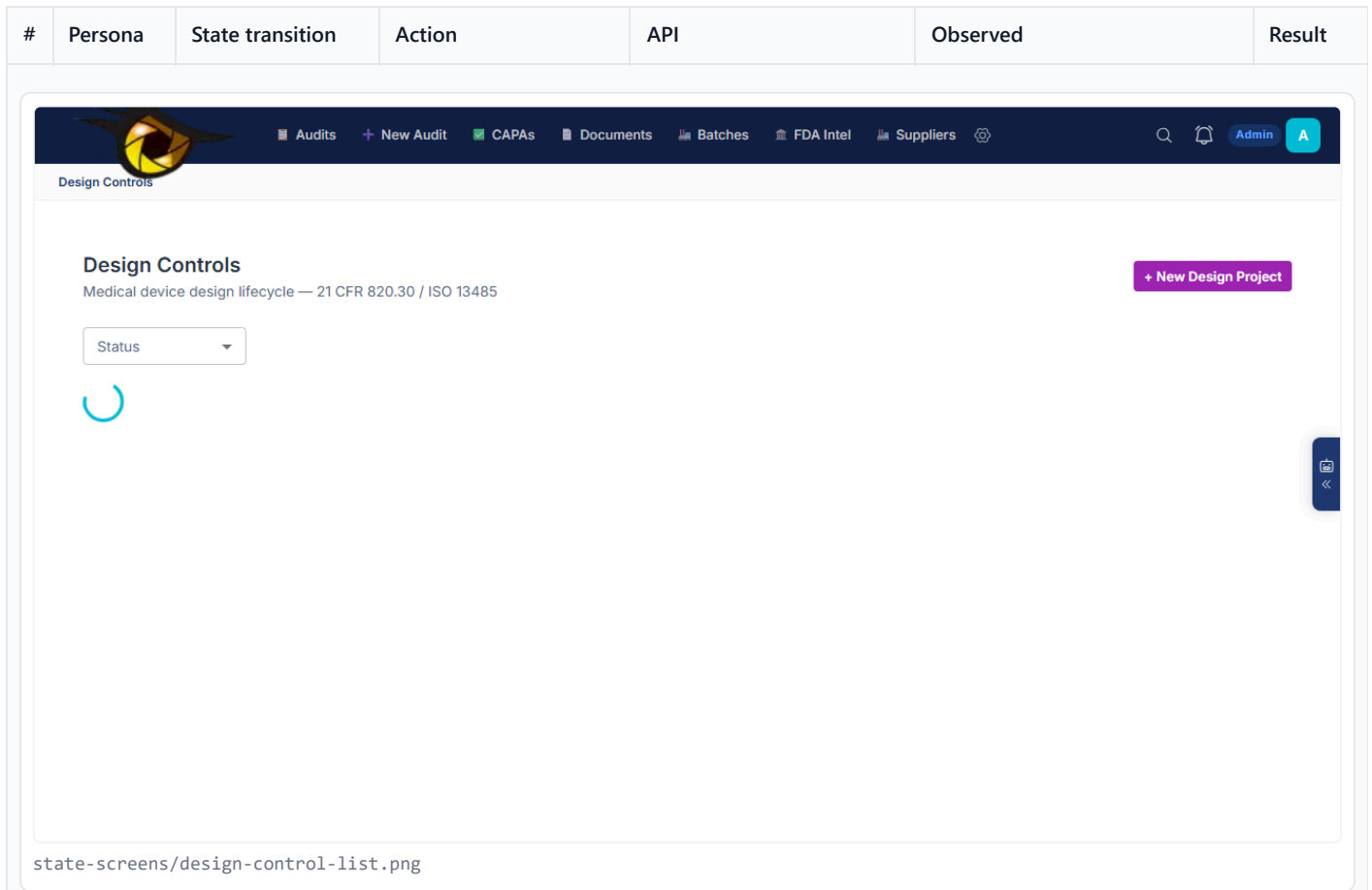
5. Full lifecycle walkthrough — one transaction, end-to-end

One Class II 510(k) device walked from PLANNING through TRANSFER.

#	Persona	State transition	Action	API	Observed	Result
1	Product Engineer engineer	– → DRAFT	+ Start Design · CLASS_II · 510K	POST /api/design-controls	designNumber=DC-2026-NNNN · status=DRAFT · 8 phases initialized	PASS

Expected DB state after this step:

```
design-controls { _id, designNumber, deviceClass: 'CLASS_II', regulatoryPathway: '510K', currentPhase: 'PLANNING', phases: [8 entries with status NOT_STARTED], status: 'DRAFT' }
```



2	Engineer engineer	DRAFT → ACTIVE	Advance phase (PLANNING starts) → status auto-flips ACTIVE	POST /api/design- controls/:id/advance- phase	Status=ACTIVE · phases[0].status=IN_PROGRESS	PASS
3	Engineer engineer	ACTIVE (PLANNING) → ACTIVE (INPUT)	Capture design inputs · Advance phase	POST /api/design- controls/:id/advance- phase	phases[0]=COMPLETED · phases[1]=IN_PROGRESS	PASS
4	Engineer engineer	ACTIVE (INPUT) → ACTIVE (OUTPUT)	Add design outputs traceable to inputs · Advance	POST /api/design- controls/:id/advance- phase	phases[1]=COMPLETED · phases[2]=IN_PROGRESS	PASS
5	James QA reviewer	ACTIVE (REVIEW) → ACTIVE (VERIFICATION)	Add design review · decision=PROCEED · Advance	POST /api/design- controls/:id/reviews + advance-phase	designReviews[] populated · phase advances	PASS
6	James QA	ACTIVE (VERIFICATION) → ACTIVE (VALIDATION)	Attach verificationProtocolRef + verificationReportRef · Advance	PUT + advance-phase	Phase advances	PASS

#	Persona	State transition	Action	API	Observed	Result
7	Marcus Reg Affairs	ACTIVE (VALIDATION) → ACTIVE (TRANSFER)	Attach validation report · Advance	PUT + advance-phase	Phase advances	PASS
8	Marcus + James RA + QA	ACTIVE (TRANSFER) → TRANSFERRED	POST /transfer · manufacturingSiteId	POST /api/design- controls/:id/transfer	Status=TRANSFERRED · currentPhase=CHANGES · DHF locked	PASS

Expected DB state after this step:

```
design-controls { status: 'TRANSFERRED', currentPhase: 'CHANGES', dhfRef }
```

6. AI assistance map

Each AI assist is a separate endpoint. Outputs flow through the grounded-generation runtime (citation gate, confidence floor, schema validation, accept/reject audit trail).

AI agent	Attached to state(s)	Endpoint	Where in UI	What it returns	Provider
(roadmap) AI design-input traceability checker	ACTIVE (INPUT/OUTPUT)	(future)	Inputs / Outputs tabs	Cross-check that every input has at least one output that traces to it; flag orphans	rule + Free Gemini for narrative

7. Regulator-trace matrix

For each lifecycle state, the specific clauses + evidence Hawkeye captures so an inspector can audit the record. Maps to FDA / ISO / ICH / EU GMP source clauses.

State	Cited clauses	Evidence captured	Backing records
ACTIVE (PLANNING)	21 CFR 820.30(b)	Design plan + responsibilities + reviews schedule	design-controls
ACTIVE (INPUT)	21 CFR 820.30(c)	designInputs[] with source + priority + verified flag	design-controls
ACTIVE (OUTPUT)	21 CFR 820.30(d)	designOutputs[] with documentRef + tracedToInput	design-controls
ACTIVE (REVIEW)	21 CFR 820.30(e)	designReviews[] with attendees + decision + minutesRef	design-controls
ACTIVE (VERIFICATION)	21 CFR 820.30(f)	verificationProtocolRef + verificationReportRef	design-controls
ACTIVE (VALIDATION)	21 CFR 820.30(g)	validationProtocolRef + validationReportRef	design-controls

State	Cited clauses	Evidence captured	Backing records
TRANSFERRED	21 CFR 820.30(h) 21 CFR 820.30(j)	manufacturingSiteId + dhfRef + transferredAt	design-controls

8. Test results matrix

Suite	Scope	Result	Evidence
(deferred) eqms-lifecycle.spec.ts · design-control	Full 8-phase walkthrough	SKIP	Not yet automated end-to-end

9. Known gaps + roadmap

- **Full 8-phase E2E lifecycle test** — Engineer + QA + RA + Manufacturing orchestration spec. **MEDIUM**
- **AI traceability checker** — Cross-check inputs ↔ outputs ↔ verification ↔ validation chain. **MEDIUM**
- **DHF auto-assembler on TRANSFER** — Auto-bundle all phase artifacts into a single PDF DHF on transfer. **HIGH**
- **ISO 14971 risk-management plan integration** — linkedRiskItems[] should auto-pull from risk-register at PLANNING phase start. **MEDIUM**